ANHAD SINGH SARNA

DEVOPS ENGINEERING / CLOUD & INFRASTRUCTURE AUTOMATION / RELIABILITY

Edmonton, Alberta • (780) 729-2480 • assarna@ualberta.ca • github.com/anhads3ngh

PROFESSIONAL PROFILE

DevOps-focused software engineer and University of Alberta Computer Science student (3.87 GPA) with enterprise experience in Python automation, Jenkins CI workflows, API integration, log-driven troubleshooting, and Agile software delivery. Built PulseOps, a cloud-native reliability project featuring AWS infrastructure as code, Docker, GitHub Actions, SQS retries, dead-letter queues, and monitoring. Communicates proactively, takes ownership of technical issues, and collaborates across engineering teams.

TECHNICAL QUALIFICATIONS

Cloud & IaC AWS (Lambda, SQS, DynamoDB, CloudWatch, IAM, ECR, API Gateway); Terraform; GitHub OIDC (PulseOps infrastructure configuration)

CI/CD & platforms Jenkins, GitHub Actions, Git, Bitbucket, Docker, Docker Compose, Linux, Bash, container image builds, Trivy

Development Python, SQL, C/C++, C#, FastAPI, Flask, REST APIs, JSON, pytest, Ruff, Selenium, Appium, REST Assured, Postman

SRE & security Observability, structured logs, CloudWatch alarms, queue depth, retry/backoff patterns, idempotency, dead-letter queues, IAM least privilege

Delivery & teamwork Agile/Scrum, JIRA, code feedback, root-cause analysis, regression automation, release readiness, technical documentation

ENGINEERING EXPERIENCE

TD Bank | Software Engineering Intern — DevOps & Automation Focus — Toronto, ON | Jan–Aug 2026

- Built Python-driven API, web, and mobile verification workflows using Selenium, Appium, and REST Assured; connected automated checks to Jenkins-based continuous integration practices.
- Managed automation scripts and test assets with Git and Bitbucket; collaborated with engineers on incremental code changes, review feedback, reproducible defects, and Agile sprint delivery.
- Diagnosed integration and reliability issues by correlating application logs, SQL queries, REST API responses, and end-to-end system behavior; documented root-cause evidence and next steps.
- Supported release validation and operational readiness through repeatable checks, defect triage, and concise engineering handoffs across development and product teams.
- Prototyped Python and LangChain-assisted automation for testing and information-retrieval workflows, exploring ways to reduce repetitive manual analysis.

University of Alberta | Computing Science Teaching Assistant — Edmonton, AB | Jan 2024–Present

- Lead Python-focused labs and mentoring sessions; coach students through debugging, algorithms, data structures, program design, and systematic problem-solving.
- Review code for logic, edge cases, readability, and quality; explain improvements through actionable feedback and accessible technical examples.
- Previously served as Head TA for CMPUT 174, coordinating student support and sharing consistent learning resources with other course staff.

Space Copy | Software Development Intern — 2025

- Contributed to software development at an aviation and space-additive-manufacturing startup, navigating evolving requirements and collaborating on technical problem-solving.

EDUCATION

University of Alberta — BSc in Computer Science (Artificial Intelligence) | Expected May 2027 | GPA 3.87

Relevant coursework: Operating Systems, Computer Security, Software Engineering, Databases, Data Structures & Algorithms, Artificial Intelligence

FEATURED DEVOPS AND OBSERVABILITY PROJECTS

PulseOps — Cloud-Native Delivery Reliability Platform | Python · FastAPI · AWS · Terraform · Docker · GitHub Actions

Created an event-driven simulated social-post delivery service with an API, dashboard, SQS jobs, DynamoDB state, and Lambda-compatible workers. Implemented idempotency, worker leases, transient retries, DLQ handling, and partial-batch failures; authored AWS Terraform, least-privilege/OIDC deployment configuration, Docker Compose and CloudWatch telemetry. Verified 10 offline unit tests and documented the architecture with a video walkthrough.

Distributed Cassandra Log Explorer | Python · SQLite · Docker Compose · OpenSearch REST API · JavaScript

Built a local observability prototype for synthetic logs from three database nodes; parsed structured fields and exposed searchable events with filters by node, severity and keywords. Added optional OpenSearch integration and automated parser/query tests.

SOFTWARE SYSTEMS, ALGORITHMS AND APPLICATIONS

Multi-Agent Pathfinding Engine | Python · A* · Conflict-Based Search

Implemented CBS with A* for collision-free multi-agent grid routes, including path constraints, conflict detection and deterministic test instances.

3D Ray-Tracing Renderer | C · GCC · Make · Computer Graphics

Built a renderer with vector math, ray–sphere intersections, lighting, shadows, memory management, and image/file output.

AI-Powered Aircraft Maintenance Scheduler | Python · TensorFlow · Pandas · NumPy

Developed a reinforcement-learning scheduling prototype for aircraft maintenance, balancing operational constraints, resource use and downtime.

FIFA World Cup Analytics & Predictive Modeling | Python · Scikit-learn · Power BI

Cleaned match/team datasets, trained a logistic-regression outcome model, and presented performance metrics through interactive dashboards.

Interactive 3D Adventure Prototype | Unity · C# · Blender · Substance Painter

Built a modular interactive world with C# gameplay logic, 3D assets, lighting, effects and user-focused interactions.

Sudoku Constraint-Satisfaction Solver | Python · Backtracking · MRV · AC-3

Implemented backtracking search with heuristics and AC-3 constraint propagation to solve Sudoku efficiently and analyze search behavior.

Image Classification: CNN vs. MLP | Python · PyTorch · Modified MNIST

Trained and compared convolutional and fully connected neural networks for image classification on a modified handwritten-digit dataset.

OTHER PROJECTS AND EXPLORATION

Emotion-Based Music Recommender · **Intelligent Study Companion** — Additional AI-oriented portfolio projects.